

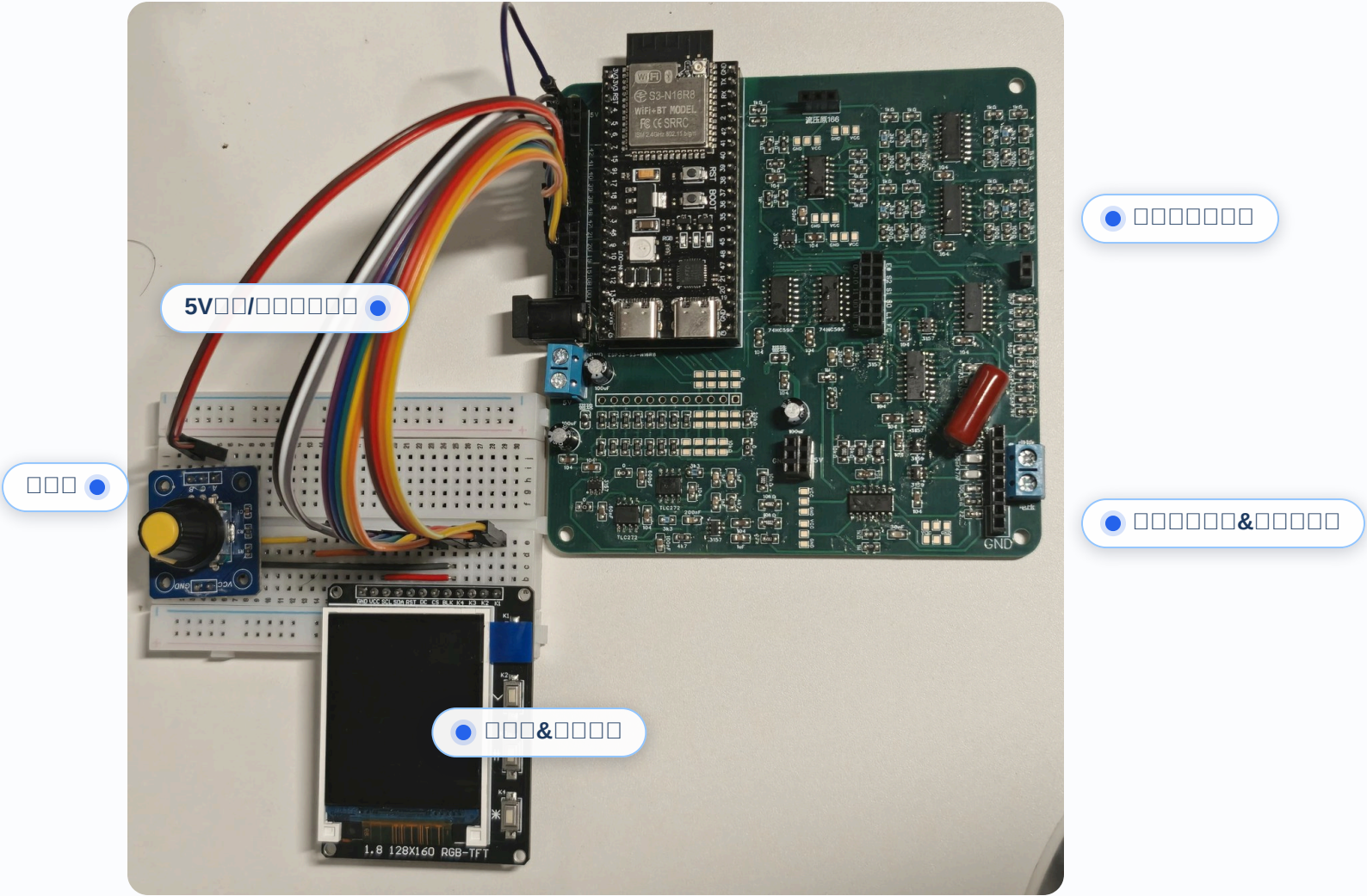
LCR ANALYZER · MULTIFUNCTIONAL LCR METER

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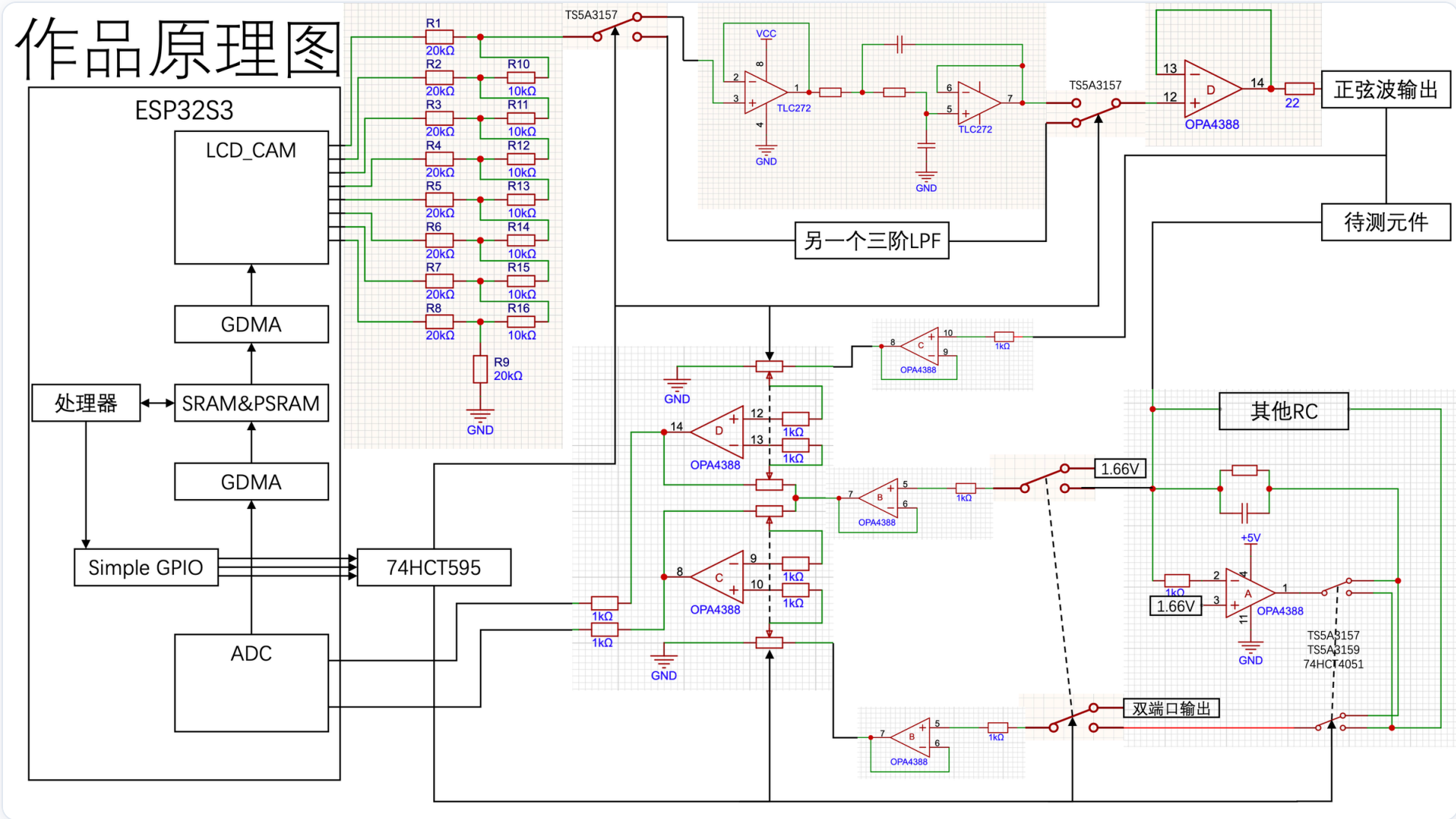
LCR ANALYZER · HARDWARE



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R□□□□□220□□-10□□

R□□□

100Hz 1k-300k:±1%

100Hz 47k-10M:±2%

□□□□□□□□□□±3%□□□

□□□□□□□□□□□□□□±1°□□□

□R□□□□□□□□□□□□□□□□

R	100Hz			1kHz			10kHz		
万用表测量值	Rtest	Deg	误差/%	Rtest	Deg	误差/%	Rtest	Deg	误差/%
10.2M	10.1M	-0.355	-0.98	10.4M	-0.267	1.96	10.4M	-6.19	1.96
3.85M	3.87M	-0.148	0.52	3.89M	0.0712	1.04	4.02M	-1.27	4.42
984k	983k	0.0646	-0.10	984k	0.433	0.00	1.03M	2.2	4.67
677k	688k	-0.0142	1.62	692k	0.0776	2.22	700k	2.25	3.40
298k	299k	-0.0192	0.34	298k	0.0468	0.00	307k	0.57	3.02
219k	220k	-0.0651	0.46	222k	0.0328	1.37	223k	0.621	1.83
99.8k	99.7k	-0.0653	-0.10	99.9k	0.0398	0.10	101k	0.304	1.20
67.5k	67.9k	0.0216	0.59	68.5k	0.186	1.48	67.8k	0.312	0.44
33.0k	32.9k	0.00527	-0.30	33k	0.00847	0.00	33.4k	0.141	1.21
20.0k	20.1k	-0.0642	0.50	20.2k	-0.0154	1.00	19.8k	0.00516	-1.00
10.0k	9.99k	-0.0582	-0.10	10k	-0.0111	0.00	10.1k	0.0237	1.00
6.73k	6.72k	0.0289	-0.15	6.77k	0.0557	0.59	6.7k	0.00555	-0.45
3.89k	3.86k	-0.0131	-0.77	3.88k	0.0278	-0.26	3.9k	0.298	0.26
2.19k	2.21k	-0.0773	0.91	2.23k	-0.00152	1.83	2.17k	0.0381	-0.91
990ohm	989ohm	-0.0558	-0.10	991ohm	-0.00336	0.10	999ohm	-0.0265	0.91
681ohm	693ohm	0.00656	1.76	697ohm	-0.0818	2.35	679ohm	-0.0638	-0.29
272ohm	273ohm	-0.0151	0.37	271ohm	0.143	-0.37	277ohm	-0.171	1.84
217ohm	213ohm	-0.0311	-1.84	219ohm	-0.0126	0.92	219ohm	-0.112	0.92
103ohm	103ohm	-0.0378	0.00	103ohm	-0.0211	0.00	103ohm	0.0587	0.00
47.9ohm	47.4ohm	-0.0566	-1.04	47.5ohm	0.0133	-0.84	47.2ohm	-0.0459	-1.46
29.9ohm	29.1ohm	-0.0364	-2.68	29.9ohm	-0.00654	0.00	29.5ohm	0.0962	-1.34
10.0ohm	10.1ohm	-0.0639	1.00	10.1ohm	-0.0313	1.00	10.1ohm	-0.089	1.00
0.22ohm	210m	-0.169	-4.55	210m	-0.35	-4.55	210m	-3.25	-4.55

注：测量值为万用表测量该电阻时的读数，保留最多三位有效数字

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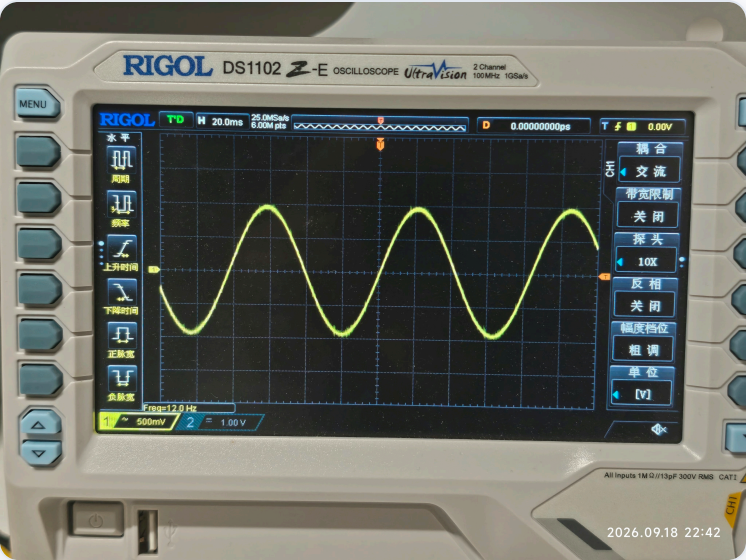
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The graph illustrates the relationship between residual capacity and frequency for different electrolyte configurations. The y-axis represents residual capacity in percent, ranging from 70 to 110. The x-axis represents frequency in Hz, with five discrete points. The legend identifies six data series: '有极性电解' (Polar Electrolyte) in dark blue, '独石1' (Ceramic 1) in orange, '独石2' (Ceramic 2) in grey, 'CBB' in yellow, '无极性电解(10u)' (Non-polar Electrolyte 10u) in light blue, and 'CBB(10u)' in green.

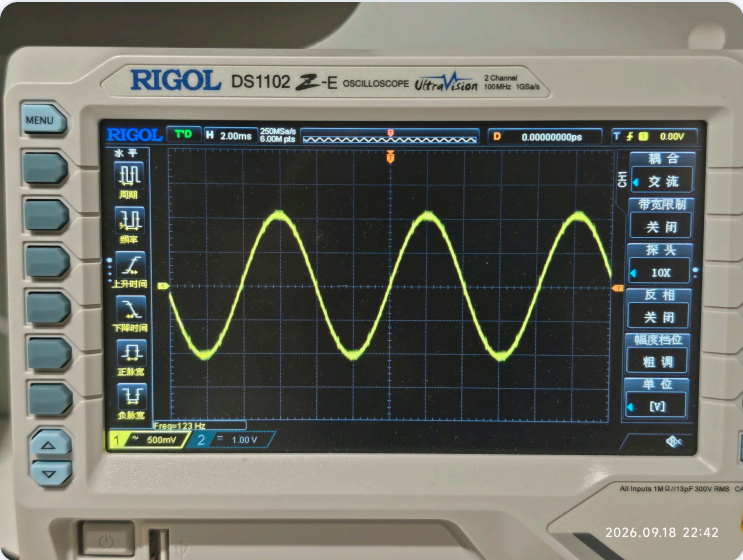
频率/Hz	有极性电解	独石1	独石2	CBB	无极性电解(10u)	CBB(10u)
100	108	83.5	98.5	101	106	100
200	107	81.5	96.5	100	99.5	100
300	103	81	95.5	100	87	101
400	99	75.5	92	101	77.5	98
500	99	75.5	92	101	77.5	98

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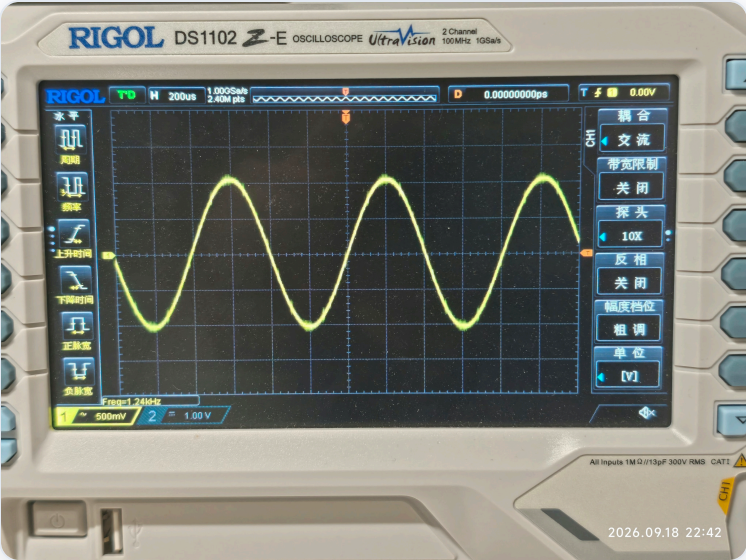
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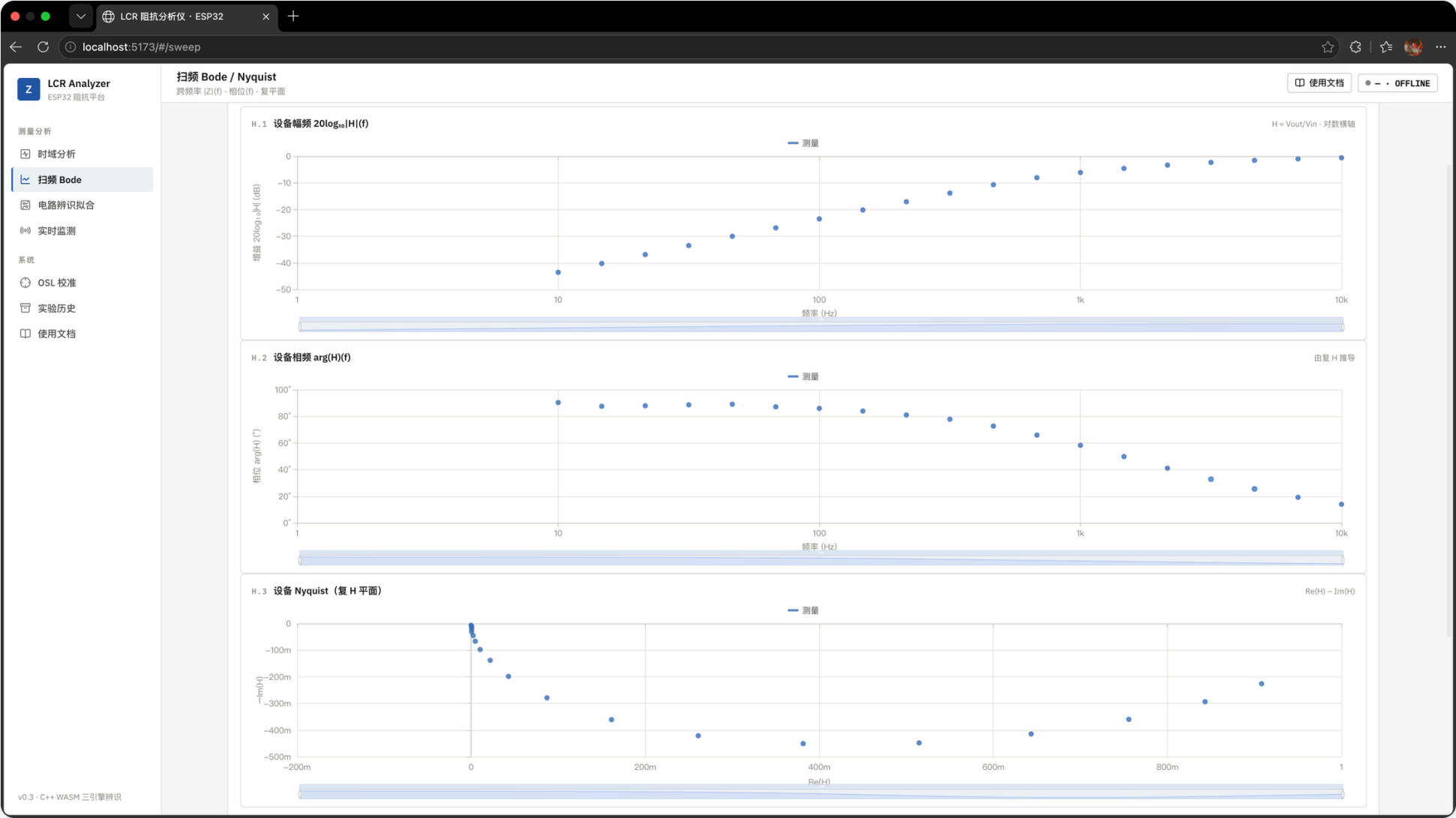
12 Hz □□□□



123 Hz □□□□



1234 Hz □□□□



LCR ☐☐☐☐☐☐☐☐☐

☐☐☐☐ · Try1–Try3☐☐☐☐☐☐☐☐☐☐☐☐☐☐☐☐☐

$$\text{□□□□□ } Z(f_k) \text{ □□□□□ } G \text{ □□□ } \theta \text{ □□ } Z_{\text{calc}}(f_k) = b^T Y^{-1}(f_k) b \text{ □□□□}$$

☐☐ AlgorithmLcr v4.1.2 ☐☐ C++/Eigen ☐☐☐☐ CLI ☐☐☐ WASM ☐☐

☐☐☐☐☐☐☐☐☐☐☐☐☐☐☐☐☐☐☐☐ 10 Hz–10 kHz☐☐☐☐



□□□ **IV**□□□□□□□□□□ $Z = b^T Y^{-1} b$

- ^[2]
- □□ 0 □□□□□□ 1 □□□□ 1 A□□□ $b = e_1$ □□□ 1 □□□□□□□□□□
 - KCL □□□□□ $Y v = b$ □□□ $v = Y^{-1} b$ □
 - □□ $I_{\text{port}} = 1$ A□□□□□□□□□□□□□□ $Z = V_1$ □

[$Z = V_1 = b^T v = b^T Y^{-1} b$]

□□□□□□ R - C □□□□
□□□□ {1,2}□ $g = 1/R$, $y_C = j\omega C$ □

$$Y = \begin{bmatrix} g & -g \\ -g & g + y_C \end{bmatrix}, \quad b = \begin{bmatrix} 1 \\ 0 \end{bmatrix} \Rightarrow Z = \frac{g + y_C}{g y_C} = R + \frac{1}{j\omega C} \checkmark$$

□□□ **V**□□□□□□□□□□□□□□

□□□□□□□□□□□□

$$z_{\text{floor}} = \max(10^{-15}, 10^{-9} \text{ median}|Z_k|), \quad r_k = \frac{\begin{bmatrix} \text{Re}(\hat{Z}_k - Z_k) \\ \text{Im}(\hat{Z}_k - Z_k) \end{bmatrix}}{\max(|Z_k|, z_{\text{floor}})}$$

- □□□□ $J = \sum_k \|r_k\|^2$ □□□□□□ $1\text{ k}\Omega \square 1\text{ }\Omega$ □□□□□□□□□□□□□□□□
- □□□□□□□□ **GLS**□□□□□□□□□□□□□□□□□□□□

WRMSE □□□□□□ $\text{WRMSE} = \sqrt{\frac{1}{M} \sum_{k=1}^M \ r_k\ _2^2} = \sqrt{J/M}$ □□□□	MAXREL □□□□□□ $\text{MAXREL} = \max_{1 \leq k \leq M} \ r_k\ _2$ □□□□	AICc □□□□□ [13] [14] $\text{AICc} = n \log(\max(J/n, 10^{-300})) + 2k + \frac{2k(k+1)}{n-k-1}$ $n = 2M, k = p + 1$□□□□□□□□□□□□□□□□□□□□
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□□□□□□□□□□□□**Try1** □□□□□□□□□□□□**Try2** □□□□□□□□□□□□**Try3** □□□□□□□□□□□□

$$\text{|||||} \iff 2 \leq V \leq E+1 \quad (E \geq V-1)$$

- **canonicalization** □ [16]

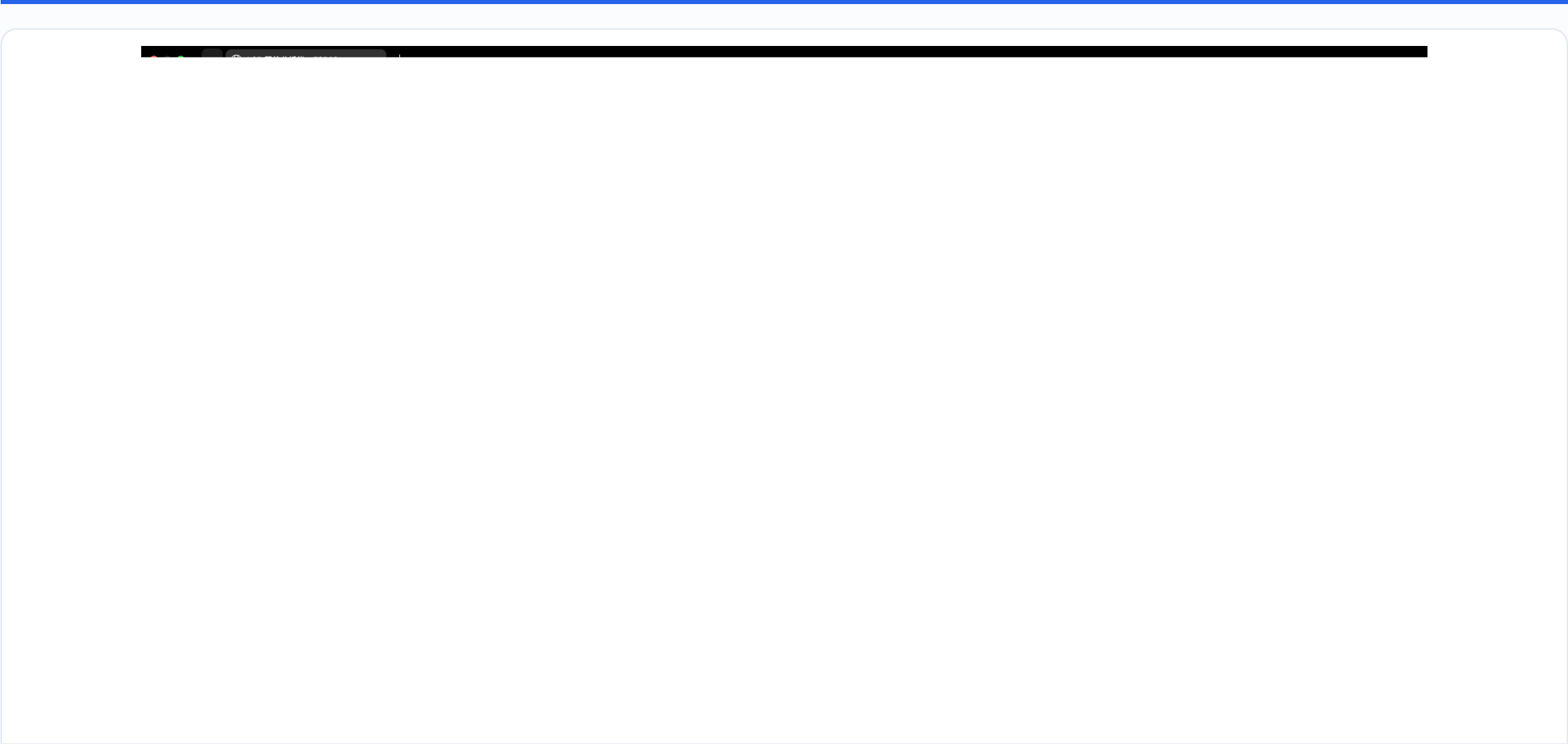
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- □□ $E < 8$ □□□□□□□□□□□□□□□□ E □□□□ $E=7$ □□□□□□□□

- 0 RSS
- Top-K 10^{-6}
- $\text{rank}(\mathbf{A}) + \text{rank}(\mathbf{B}) \Leftrightarrow \text{rank}(\mathbf{A} + \mathbf{B})$

Try1

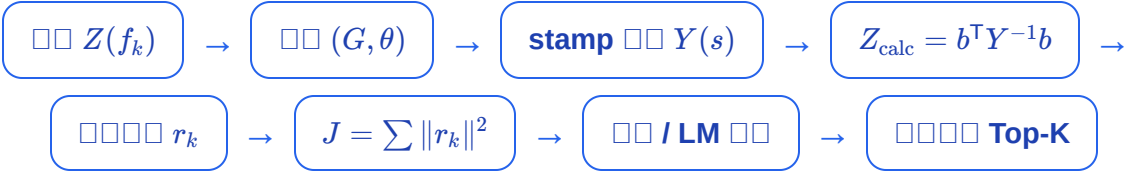


Try3 ☐☐☐☐☐☐☐☐☐☐☐☐☐☐☐



□□□□□□□ Try1-Try3

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- Try1□□□□□ SP □□□□□□□□□□□□□□□□
- Try2□□□□□□□□□□□□□□□□□□□□□□□□□□
- Try3□□□□□□□□□□□□□□□□ LM □□□□□□□□□□□□□□□□

□□□□□□□□□□□□□□□□□□□□ 1 A □□□□ → □ Yv = b → □□ Z = V₁ → □□□□□□□□

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LCR Analyzer · Try1–Try3 □□□□

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